7 **(a)** Gravitational field is <u>a region of space</u> [B1] where a <u>mass will experience a gravitational force</u> due to the presence of another mass. [B1]

(b) Define *gravitational potential* at a point is the <u>work done per unit mass</u> in bringing a small test mass from infinity to that point without a change in kinetic energy. [B1]

(c) **(i)**

$$\phi = -\frac{GM}{R}$$

$$\Delta\phi = \phi_B - \phi_A$$

$$= -\frac{GM}{6R} - \left(-\frac{GM}{2R}\right) \quad \text{[M1]}$$

$$= \frac{GM}{3R} \quad \text{[A0]}$$

(ii)

loss in E_k = Gain in E_p

$E_{k,initial}$ - $E_{k,final}$ = m$\Delta\phi$

$$(1.70 - 0.72) \times 10^{12} = 4.7 \times 10^4 \times \frac{GM}{3R}$$

$$0.98 \times 10^{12} = 4.7 \times 10^4 \times \frac{4.0 \times 10^{14}}{3R} \quad \text{[M1]}$$

$$R = 6.39 \times 10^6 \text{ m [M1]}$$

Distance A to B = 4R = 2.56 x 10^7 m [A1]

assumption: Ignoring gravitational force due to other bodies. [B1]

(iii) Gravitational force provides for centripetal force on rocket.

$$\frac{GMm}{(6R)^2} = \frac{mv^2}{6R}$$

$$v^2 = \frac{GM}{6R}$$

$$v = \sqrt{\frac{4.0 \times 10^{14}}{6 \times 6.39 \times 10^6}} \quad \text{[M1]}$$

$$= 3230 \text{ m s}^{-1} \quad \text{[A1]}$$

(d) E*lectric field strength* at a point is the electric force per unit positive charge placed at that point. [B1]

(e) Similarity [B1 for any of the following]
1) Both field lines are radial from the centre of the sphere.
2) Both field lines intersects perpendicular to the surface of the sphere.
3) Field lines are equally spaced at the surface of the sphere.
4) Further from the sphere, the spacing of lines increases (indicating that field strength becomes weaker away from charge)

Difference [B1]
1) Gravitational field lines are directed towards centre of sphere while electric field lines are directed away from the centre of sphere.

(f) **(i)**

$$E = \frac{Q}{4\pi\varepsilon_0 r^2} \quad \text{[C1]}$$

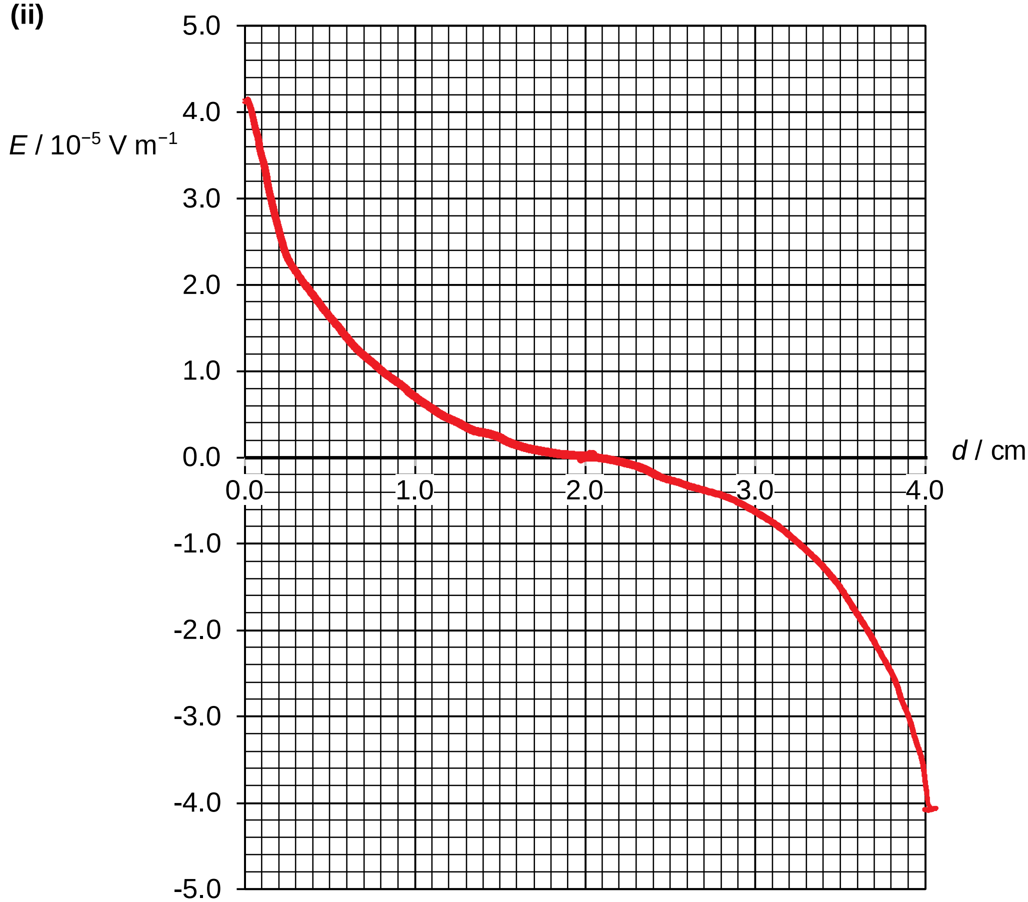
$$E \text{ at A} = E_1 - E_2$$

$$4.1\times10^{-5} = \frac{Q}{4\pi\varepsilon_0 r_1^{\,2}} - \frac{Q}{4\pi\varepsilon_0 r_2^{\,2}}$$

$$= \frac{Q}{4\pi\varepsilon_0 (0.01)^2} - \frac{Q}{4\pi\varepsilon_0 (0.05)^2} \quad \text{[M1]}$$

$$Q = 4.75\times10^{-19}\text{ C} \qquad \text{[A1]}$$

(ii)



[B1] smooth curve & generally asymmetrical about horizontal axis

[B2] zero E at d= 2.0 cm, correct intercepts at 4.1 x 10^{-5} & -4.1 x 10^{-5}

(iii) By Newton 2nd law, F_{net} = qE = ma, the acceleration of the charge is proportional to the electric field strength.

Hence, from A to midway between A & B, the charge accelerates (towards the right) at a decreasing rate until it reaches zero at midway. [B1]

Thereafter, the charge accelerates (in opposite direction) from zero at an increasing rate towards B. [B1]

Alternative answer to 2nd mark: Thereafter, the charge decelerates at an increasing rate towards B.